

Inpred production:
BioInformatics QC pipeline

Standard Operation Procedure

A new patient has successfully been sequenced:

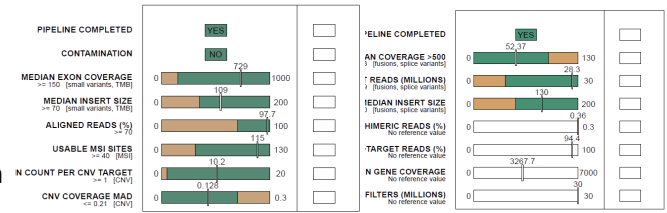
1. TSO500 LoaclAPP from Illumina
2. "TSOPPI" (TSO500 Post-Processing for InPreD): In-house post processing (includes PCGR and CPSR)



QC of the sample



If a QC metric fails, contact the lab.
NB: If only CN-MAD fails we write a comment.



Illumina threshold

B-allele frequency plot: AF of SOM and GL_DB variants. Inspect CN variation, LOH, T%.



Comment if the data suggests underestimated T%

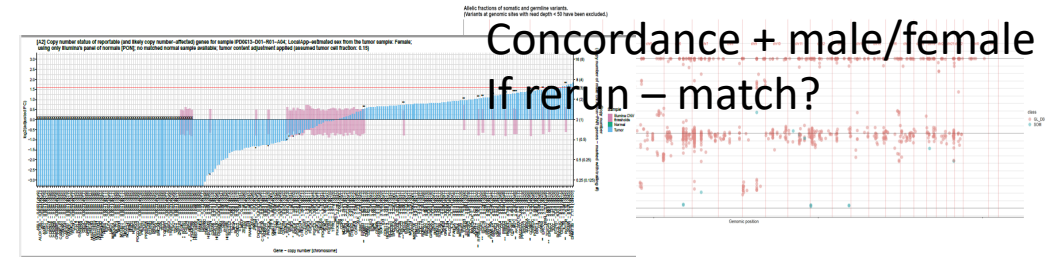
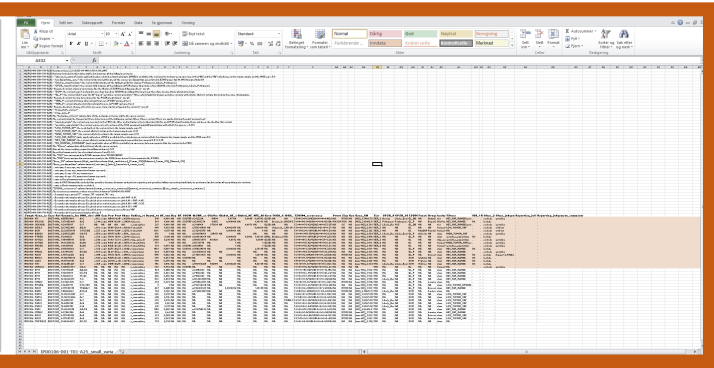


Table for
Medisinsk Genetikk:
variants in cancer
predisposition genes
(Mandelker et al. 2019,
ESMO recommendations)

Variant inspection:

- (Possible) Germline variants
- Background noise (artifact)
- Low coverage
- Low AF
- Alignment error (IGV)



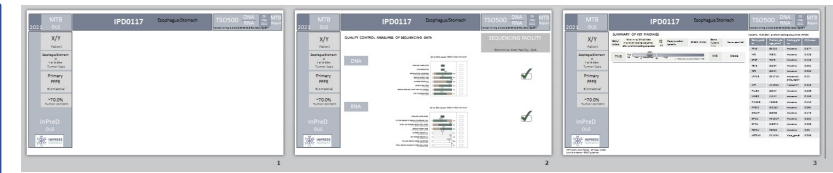
**Compare findings
to ctDNA:**
Comment on
discrepancies and
added biomarkers

QC:
Bioinformatics

Automatic

Manual

Automatically print the initial report for Mol-MDT.
IPD id info, T%, date, versions. QC plots,
MSI, TMB, number of variants, table with non-syn.



QC reporting: RNA (fusion, splicing) and CN (gain, loss)

Small variant table

File

Hjem

Sett inn

Sideoppsett

Formler

Data

Se gjennom

Visning

Klipp ut

Kopier

Lim inn

Kopier format

Utklippstavle

Arial

10

Bryt tekst

Standard

Betinget formatering

Normal

Dårlig

God

Nøytral

Beregning

Forklarende...

Inndata

Koblet celle

Kontrollcelle

Merknad

Sett inn

Slett

Format

Celler

Σ Autosummer

Fyll

Fjern

Sorter og Søket etter filterer

Redigering

Utklippstavle

Skrift

Justering

Tall

Stiler

IPD0041-D01-C01-F24_small_variant_table_t0cand0r.v1.xls

[kompatibilitetsmodus] - Microsoft Excel

B

C

D

E

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H

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J

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P

Q

R

S

T

U

V

X

Y

Z

AA

AB

AC

AD

AE

AF

AG

AH

AI

AJ

AK

AL

AM

AN

AO

AP

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100

Version string: 2.2.0.1221-03-21/0.9.1/0.6.1/hg19T

Variables included in this table satisfied at least one of the following criteria:

- "absent_in_normal": (only applicable when a matched normal sample [MNS] is available) the variant site had non-zero coverage in the MNS and the VAF ratio between the tumor sample and the MNS was >=0.5

- "predisposition_gene": the variant is located within one of the cancer predisposition genes listed in ESMO paper doi:10.1093/annonc/mdz136

- "ClinVar_classification": the variant falls into one of the following ClinVar classes: Pathogenic, Likely_Pathogenic

- "CPSR_classification": the variant falls into one of the following CPSR classes (based on refined ACMG criteria): Pathogenic, Likely_Pathogenic

- Required variant class (as determined by the Illumina TSO500 Local App pipeline) - any of:

- "SOM": the variant wasn't included in any class based on TSO500 Local App filtering, it was therefore deemed to be of somatic origin

- "GL_P": the variant didn't pass the AF-based "germline variant proximity" filter, which looks for known germline variants with similar allelic fractions located on the same chromosome

- Required variant tier (as determined by the PCGR annotation) - any of:

- "TIER_1": variant of strong clinical significance (a PCGR-assigned tier)

- "TIER_2": variant of potential clinical significance (a PCGR-assigned tier)

- Required protein change effect (in any gene transcript overlapping the variant) - any of:

- "frameshift_variant"

- "stop_gained"

The "inclusion_criteria" column lists all the inclusion criteria satisfied by given variant.

The variants might be flagged as filtered due to any of the following custom filters (these variant filters are applied during the post-processing):

- "normal_origin": the variant was seen only in the MNS; the filter and inclusion criteria (besides ClinVar and CPSR classification) have not been checked for this variant

- "germline_population": the variant was seen in at least one of the 1000 genomes/gnomAD populations with allelic frequency >=0.001

- "LOW_TUMOR_DP": the read depth at the variant site in the tumor sample was <10

- "LOW_TUMOR_VAF": the variant allelic fraction in the tumor sample was <0.03

- "HIGH_TUMOR_VAF": the variant allelic fraction in the tumor sample was >0.98

- "LOW_VAF_RATIO": (only applicable when a MNS is available) the ratio between variant allelic fractions in the tumor sample and the MNS was <5.0

- "HET_VAF_RANGE": the variant allelic fraction in the tumor sample was within the range of 0.5 - 0.15

- "NO_NORMAL_COVERAGE": (only applicable when a MNS is available) no coverage data was reported for the variant in the MNS

The "Filters" column lists all the filters failed by given variant.

Size of the target coding region (in millions of bases): 1.27

Specified tumor purity (as a fraction between 0 and 1): 1

The "TNC" tier corresponds to PCGR-assigned tier "NONCODING"

The "TNA" tier is assigned to variants seen only in the MNS (these haven't been annotated by PCGR).

"Gene_CN" column format: [High_confidence|non-High_confidence][Tumor_CN][Adjusted_Tumor_CN][Normal_CN]

"Gene_predisposition" column format: [category][note][associated_tumor_type]

- category 1: any age, any tumor type

- category 2: any age, associated tumor type only

- category 3: age < 30, any tumor type

- category 4: age < 30, associated tumor type only

- note a: Renal tumours to be excluded.

- note b: MUTYH should be included for germline-focussed tumour analysis but reporting and germline follow-up testing should only be performed on detection of two pathogenic variants.

- note c: Brain tumours to be excluded.

"TSO500_recurrence" column format: [tumor_recurrence_summary][normal_recurrence_summary][any_sample_recurrence_summary]

The recurrence summary values are written in format "X:A-B+C+D/M/N"

- X: sample type, one of "T" = tumor, "N" = normal, "A" = any

- A: number of samples of type X in which given variant was seen with VAF < 0.01

- B: number of samples of type X in which given variant was seen with 0.01 <= VAF < 0.05

- C: number of samples of type X in which given variant was seen with 0.05 <= VAF < 0.35

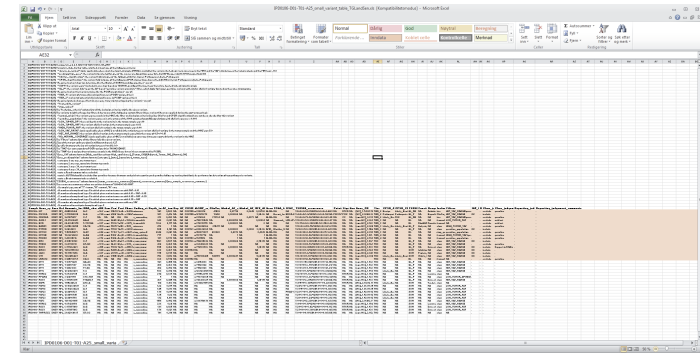
- D: number of samples of type X in which given variant was seen with 0.35 <= VAF

- M: number of samples of type X in which given variant was seen with any VAF

- N: number of investigated samples of type X

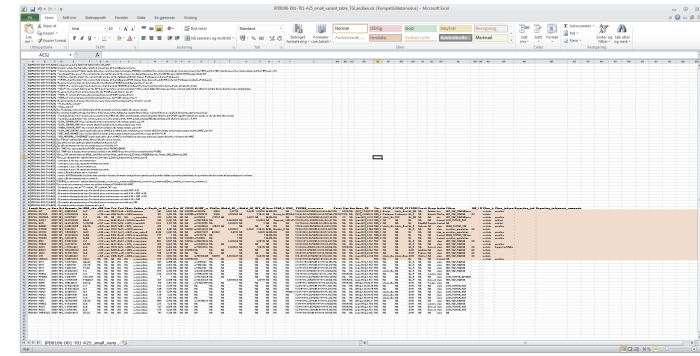
Gene_symb Ensembl RefSeq Genomic Locat DIA_change CDS1_c Exon_r Protei Protei Change_s Coding_sta Depth_tu AF_tumr Depth AF_nc COSMIC dbS ClinVar_var Global_AF_g Global_AF_NFE AF

Variants in the “small variant table”
 satisfy at least one of the following
 criteria:



- Absent in normal (applicable when a matched normal sample is available)
- Pathogenic or Likely Pathogenic (determined by either ClinVar or CPSR variant pathogenicity classification)
- Clinical significance (provided by the PCGR)
 - TIER1: strong clinical significance
 - TIER2: potential clinical significance
- Variant classification (derived from the Illumina LocalApp pipeline output)
 - GL_P: germline variant proximity class (assigned in case 5 or more germline variants with similar AFs are observed by the LocalApp on the same chromosome)
 - SOM: variants with no filter, no germline database status and no germline proximity status applied by the LocalApp
- Required protein change effect (changes generally considered to be of “HIGH IMPACT” by Ensembl)
 - frameshift_variant
 - stop_gained
 - start_lost
 - stop_lost
 - splice_acceptor_variant or splice_donor_variant

Small variant table: Include/exclude variant judgement



The screenshot shows a spreadsheet application with a table of variant data. The table has multiple columns, including 'Variant ID', 'Position', 'Class_judgement', and 'Class_judgement_comments'. The 'Class_judgement' column contains values like 'include' and 'exclude'. The 'Class_judgement_comments' column contains text descriptions of variant exclusions. The table is filtered to show only rows where 'Class_judgement' is 'exclude'.

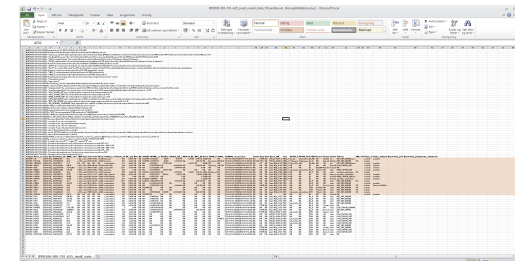
Variants passing all quality checks should be assigned value “include” in the "Class_judgement" column. Remaining variants should be assigned "Class_judgement" value “exclude”. In the latter case, comments regarding a variant’s exclusion can be recorded in column "Class_judgement_comments”.

Both above-mentioned columns are pre-filled by TSOPPI based on rules listed in the small variant table header. The pre-filling is intended to speed up the manual QC process by automating multiple well-defined quality checks, but variant reviewers should always check whether they agree with the auto-generated judgement information and conclusions.

Variant table

[illegible]

Germline variant filters



A screenshot of a genomic data table, likely from a variant caller. The table has multiple columns, including genomic coordinates, variant types, and allele frequencies. A row is highlighted in orange, indicating a specific variant of interest.

Check list:

- **Germline variants**
- Background noise (artifact)
- Low coverage
- Low AF
- Alignment error (IGV)

Tumor-sample variants are automatically flagged as likely germline variants if:

In case of a tumor-normal pair:

The variant is observed also in the normal sample ($\text{normal_AF} > 0$), and the $\text{tumor_AF} < 5 * \text{normal_AF}$. [“LOW_VAF_RATIO” filter flag]

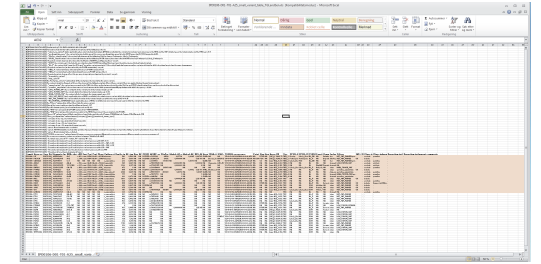
In case of tumor-only sequencing:

1. PCGR reports a 1000 Genomes/gnomAD sub-population in which the variant was observed with a frequency of at least 1/1000 (PMID: 31251990). [“germline_population” filter flag]
2. At least 10 % of the available OUS TSO500 InPreD normals contained the variant with $\text{AF} \geq 35\%$. [“recurrent_germline” filter flag]

Example

Gene symbol	DNA change		In-house recurrence (1% ; 5% ; 35%)	Illumina class	In-house observation	In-house conclusion
BARD1	CA>TG		T: 10+0+6+59=75/92 N: 0+0+0+ 21 =21/28	GL_P	Variant seen with high AF in 21 out of 28 normals	Germline
PREX2	AT>GC		T: 0+0+8+65=73/92 N:1+0+1+ 23 =25/28	GL_P	Variant seen with high AF in 23 out of 28 normals	Germline

Artifact variant filters



Position	Reference	Alternative	Quality
1000000	A	G	100
1000001	T	C	100
1000002	C	T	100
1000003	G	A	100
1000004	A	G	100
1000005	T	C	100
1000006	C	T	100
1000007	G	A	100
1000008	A	G	100
1000009	T	C	100
1000010	C	T	100
1000011	G	A	100
1000012	A	G	100
1000013	T	C	100
1000014	C	T	100
1000015	G	A	100
1000016	A	G	100
1000017	T	C	100
1000018	C	T	100
1000019	G	A	100
1000020	A	G	100
1000021	T	C	100
1000022	C	T	100
1000023	G	A	100
1000024	A	G	100
1000025	T	C	100
1000026	C	T	100
1000027	G	A	100
1000028	A	G	100
1000029	T	C	100
1000030	C	T	100
1000031	G	A	100
1000032	A	G	100
1000033	T	C	100
1000034	C	T	100
1000035	G	A	100
1000036	A	G	100
1000037	T	C	100
1000038	C	T	100
1000039	G	A	100
1000040	A	G	100
1000041	T	C	100
1000042	C	T	100
1000043	G	A	100
1000044	A	G	100
1000045	T	C	100
1000046	C	T	100
1000047	G	A	100
1000048	A	G	100
1000049	T	C	100
1000050	C	T	100
1000051	G	A	100
1000052	A	G	100
1000053	T	C	100
1000054	C	T	100
1000055	G	A	100
1000056	A	G	100
1000057	T	C	100
1000058	C	T	100
1000059	G	A	100
1000060	A	G	100
1000061	T	C	100
1000062	C	T	100
1000063	G	A	100
1000064	A	G	100
1000065	T	C	100
1000066	C	T	100
1000067	G	A	100
1000068	A	G	100
1000069	T	C	100
1000070	C	T	100
1000071	G	A	100
1000072	A	G	100
1000073	T	C	100
1000074	C	T	100
1000075	G	A	100
1000076	A	G	100
1000077	T	C	100
1000078	C	T	100
1000079	G	A	100
1000080	A	G	100
1000081	T	C	100
1000082	C	T	100
1000083	G	A	100
1000084	A	G	100
1000085	T	C	100
1000086	C	T	100
1000087	G	A	100
1000088	A	G	100
1000089	T	C	100
1000090	C	T	100
1000091	G	A	100
1000092	A	G	100
1000093	T	C	100
1000094	C	T	100
1000095	G	A	100
1000096	A	G	100
1000097	T	C	100
1000098	C	T	100
1000099	G	A	100

Check list:

- Germline variants
- **Background noise (artifact)**
- Low coverage
- Low AF
- Alignment error (IGV)

Tumor-sample variants are automatically flagged as likely artifacts if:

Based on the TSO500 recurrence information:

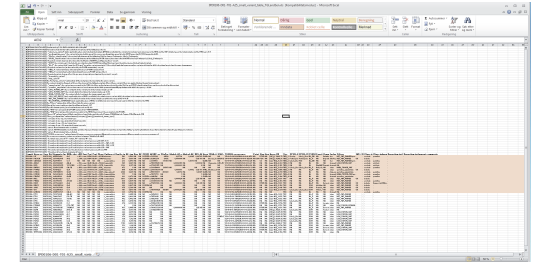
- At least 10 % of the available OUS TSO500 InPreD tumors contained the variant at AF-tier* equal or higher than the AF-tier observed in the currently processed tumor sample**.
[“recurrent_artifact” filter flag]

Please note that variants located within LocalApp’s blacklisted regions (variants classified as “BL”) may intentionally pass TSOPPI filtering if they satisfy the filter described above.

* AF-tiers: $AF < 1 \%$, $1 \% \leq AF < 5 \%$, $5 \% \leq AF < 35 \%$, $AF \geq 35 \%$

** Only variants not failing the “germline_population” filter are being checked.

Variants at low read depth locations



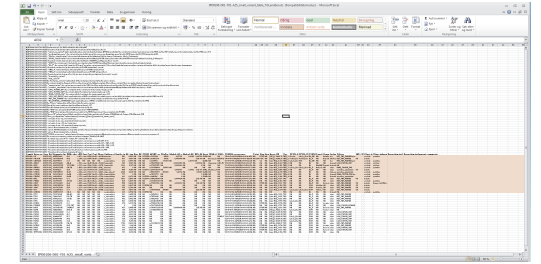
Check list:

- Germline variants
- Background noise (artifact)
- **Low coverage**
- Low AF
- Alignment error (IGV)

Tumor-sample variants are automatically flagged with the “LOW_TUMOR_DP” filter flag whenever the read depth reported for their location (by the LocalApp) is below 20.

Please note: The whitelist table output by TSOPPI provides information about selected hotspots with treatment relevance. This table can contain information about support for variants that wouldn’t be reported otherwise due to various quality issues, including read depth (e.g., TERT promoter variants).

Variants with low AF support



Check list:

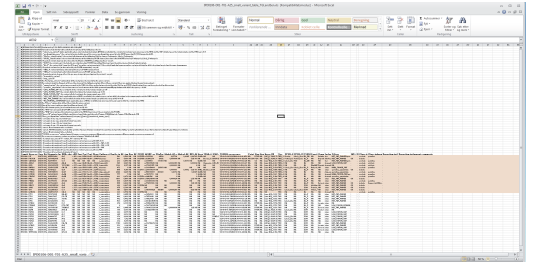
- Germline variants
- Background noise (artifact)
- Low coverage
- **Low AF**
- Alignment error (IGV)

Tumor-sample variants are automatically flagged with the “LOW_TUMOR_VAF” filter flag whenever their AF support (as reported by the LocalApp) is below 2 %.

Please note:

- The whitelist table output by TSOPPI provides information about selected hotspots with treatment relevance. This table can contain information about support for variants that wouldn't be reported otherwise due to various quality issues, including low AF.
- Small variants with $AF < 5\%$ are excluded from InPreD's TMB calculation.
- In some scenarios, such as apparent low tumor content in the sequenced sample, manual inspection might be desirable even for low-AF variants (especially in case of variants potentially relevant for treatment).

Remaining variants



Check list:

- Germline variants
- Background noise (artifact)
- Low coverage
- Low AF
- **Alignment error (IGV)**

All remaining variants are manually inspected in IGV.

Exclude based on criteria presented in paper PMID: 31779674.

Important: As this process is very time-consuming, we only inspect non-synonymous variants in IGV.